$$
\begin{aligned}
\sigma_z|u\} &= |u\} \\
\sigma_z|d\} &= -|d\} \\
\sigma_x|u\} &= |d\} \\
\sigma_x|d\} &= |u\} \\
\sigma_y|u\} &= i|d\} \\
\sigma_y|d\} &= -i|u\}. \qquad (6.6)
\end{aligned}
$$

Of course, Bob's setup is identical to Alice's, so we can write a parallel set of equations showing how the components of τ act on Bob's states: